

RecoveraX

AI Revenue Recovery Agent

Find revenue that's slipping away and win it back — safely.

RecoveraX is an AI-powered revenue recovery agent designed to identify revenue at risk, understand why the revenue was lost, determine the appropriate recovery strategy, execute bounded recovery actions, and verify whether the money was actually recovered.

Instead of blindly retrying failed payments, RecoveraX combines **AI reasoning, deterministic financial policies, Human-in-the-Loop approval, bounded retries, payment-state verification, and auditability.**

1. The Problem

Revenue can slip away through multiple failure points:

- Failed payments
- Checkout abandonment
- Failed subscriptions
- Mandate failures
- Overdue B2B invoices
- Delayed customer commitments

A simple retry mechanism treats every failure the same way.

This creates risks such as:

- Unnecessary payment retries
- Duplicate-debit risk
- Repeated failed attempts

- Poor customer experience
- Manual operational overhead
- No clear measurement of actual recovered revenue

The real problem is not just:

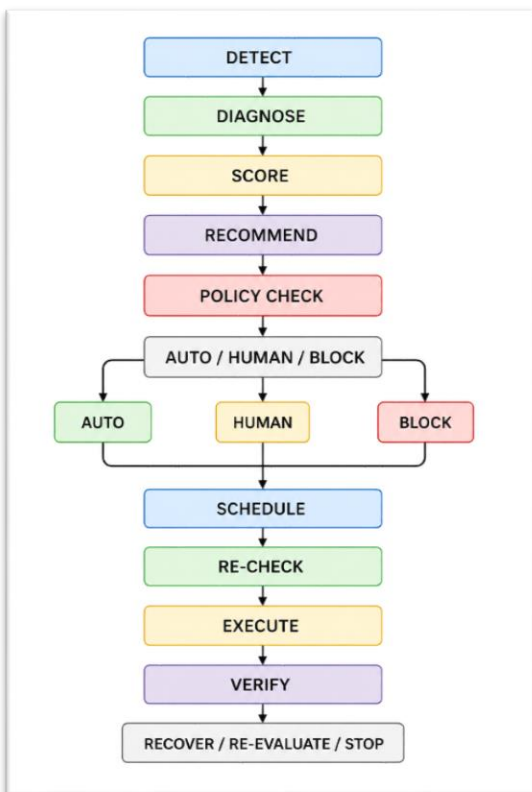
"Why did the payment fail?"

It is:

"What should we do next, is it safe to do automatically, when should we do it, and did the recovery actually succeed?"

2. Our Solution

RecoveraX closes the complete recovery loop:



Core principle

AI recommends. Deterministic policy authorizes. Executor acts. Verifier confirms.

The AI is never given direct authority to execute financial actions.

3. Why We Use AI + Deterministic Rules

RecoveraX deliberately separates **reasoning** from **financial authorization**.

AI is responsible for:

- Understanding payment context
- Diagnosing failure causes
- Reasoning over customer/payment history
- Recommending recovery strategies
- Generating recovery communication

Deterministic systems are responsible for:

- Recovery scoring
- Expected Recovery Value
- Financial authorization
- Retry limits
- Risk thresholds
- Human approval requirements
- Blocking unsafe actions
- Payment-state verification

This prevents an AI model from independently deciding to perform a potentially unsafe financial action.

4. LangGraph Agent Workflow

RecoveraX is implemented as a **stateful cyclic LangGraph workflow**.

1. Load Context

The agent loads:

- Transaction information
- Payment failure details
- Customer history
- Customer LTV
- Previous payment behavior
- Current payment state

2. Diagnose

The Groq/Qwen LLM analyzes the context and identifies the likely failure cause.

Examples:

TEMPORARY_BANK_ERROR

INSUFFICIENT_FUNDS

CARD_EXPIRED

POSSIBLE_CUSTOMER_DEBIT

3. Calculate Recovery Score

A deterministic recovery score from **0–100** is calculated using factors such as customer history, LTV, diagnosis, and recency.

Expected Recovery Value is also calculated:

EV =

Amount at Risk × Recovery Probability – Recovery Costs

4. Recommend Action

The agent recommends:

RETRY

REMIND

ESCALATE

STOP

5. Policy Check

The deterministic safety engine evaluates whether the recommended action is actually allowed.

The result is:

AUTO

HUMAN

BLOCK / STOP

5. AUTO / HUMAN / BLOCK

AUTO — Safe to automate

Low-risk and sufficiently confident cases can proceed automatically.

Example:

₹2,000

Temporary payment failure

High recovery score

No safety issue

↓

AUTO

↓

Retry

HUMAN — Human approval required

High-value, uncertain, or lower-confidence cases are routed to the merchant/operator.

Example:

₹15,000

Low confidence



HUMAN



Merchant Approval



Recovery

The workflow pauses until the required approval is provided.

BLOCK — Unsafe to continue

Ambiguous or potentially dangerous payment states are stopped.

Example:

Payment status unclear

Customer may already be debited



BLOCK



No retry

Hard safety blocks cannot be bypassed by the AI.

6. Schedule → Re-check → Execute → Verify

This is an important part of the recovery workflow.

Schedule

Once a recovery action is authorized, the system schedules it through the background worker queue.

AUTO



SCHEDULE



Wait for appropriate execution time

Re-check

Before executing the scheduled retry, RecoveraX performs a fresh payment-state check.

RE-CHECK



Already paid?



YES NO



STOP EXECUTE

This protects against unnecessary retries when the customer has already completed the payment.

Execute

The authorized recovery attempt is dispatched.

Verify

The system checks the settlement result.

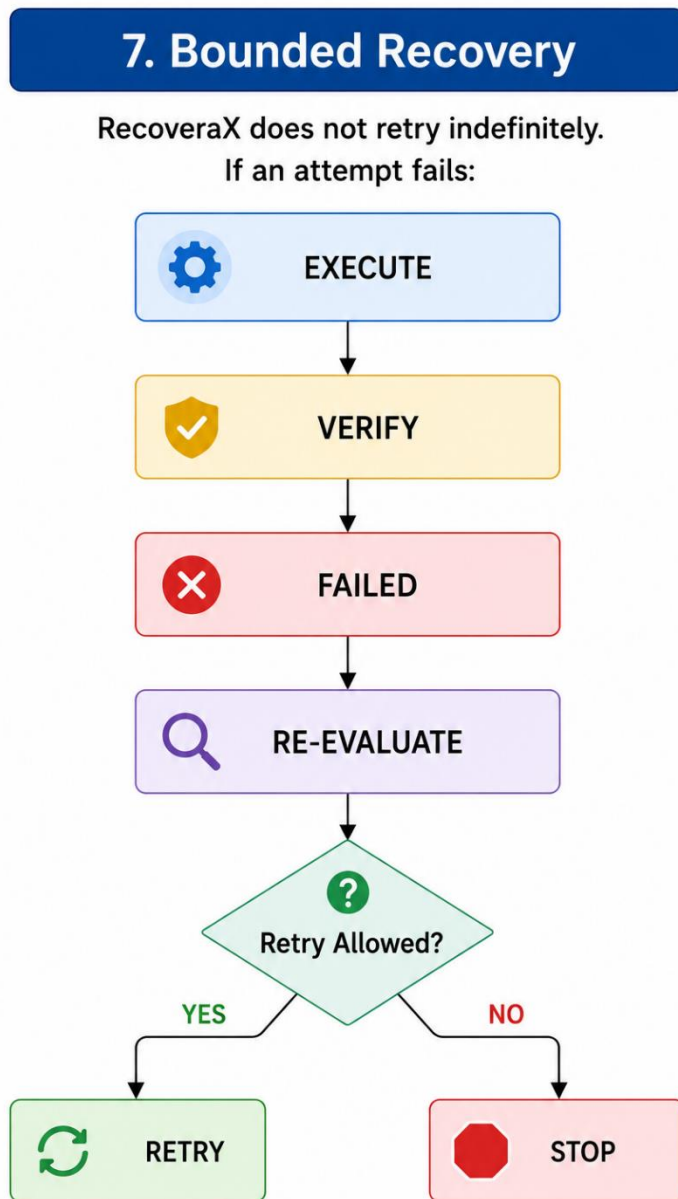
An attempted retry is not counted as recovered revenue.

Only verified payment success is counted as recovery.

7. Bounded Recovery

RecoveraX does not retry indefinitely.

If an attempt fails:



The system enforces maximum retry limits and safety conditions.

This creates **bounded autonomy instead of uncontrolled automation**.

8. Supported Revenue Recovery Scenarios

RecoveraX supports multiple Track 3 recovery scenarios.

Payment Degradation

Payment Failure



Root Cause Diagnosis



Recovery Strategy



Safe Recovery



Verification

Checkout Drop-off Recovery

Detects abandoned checkout sessions and initiates an appropriate recovery intervention to bring the customer back toward payment completion.

Failed Subscription Recovery

Handles recurring payment failures involving cards and recurring mandates, with controlled retry and reminder strategies.

B2B Receivables Chaser

Handles overdue invoices and tracks customer payment commitments.

Overdue Invoice



Customer Follow-up



Promise-to-Pay



Settlement Verification

Mandate Retry Sequencer

Supports recurring payment methods such as:

- NACH
- E-Mandate
- UPI Autopay

Retry timing incorporates cooldown and presentation-window rules instead of blindly retrying.

Hinglish Voice Recovery

Generates personalized Hinglish payment-recovery communication using Sarvam AI.

Recovery Case



Hinglish Script



Voice Generation



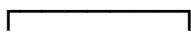
Customer Intervention

Voice intervention is restricted for unsafe or blocked cases.

Promise-to-Pay Tracker

Tracks customer commitments:

PROMISED



KEPT BROKEN

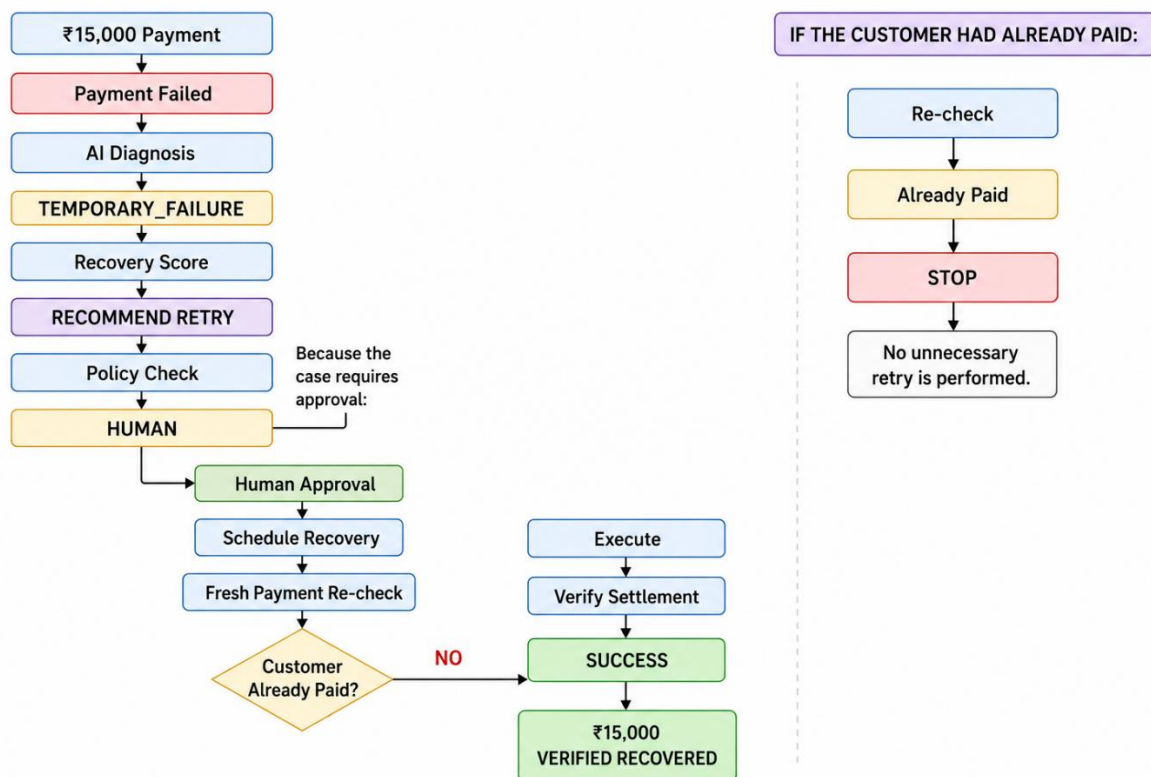
A promise is not considered recovered revenue until the actual settlement is verified.

9. Example End-to-End Case

Scenario: ₹15,000 Temporary Payment Failure

A customer attempts a ₹15,000 payment and the transaction fails because of a temporary payment issue.

RecoveraX processes the case:



10. Three Core Safety Examples

Scenario	AI Diagnosis	Action	Policy	Result
₹2K temporary failure	Temporary failure	Retry	AUTO	Verified recovery

Scenario	AI Diagnosis	Action	Policy	Result
₹15K low-confidence case	Temporary failure	Escalate	HUMAN	Approval → recovery
Ambiguous payment	Possible customer debit	Stop	BLOCK	No retry / ₹0 additional risk

11. Deterministic Safety Guardrails

RecoveraX currently applies rules including:

- **₹50,000 auto-recovery threshold**
- **Minimum recovery-score threshold**
- **Ambiguous payment → BLOCK**
- **Possible customer debit → BLOCK**
- **Fraud signal → BLOCK**
- **Maximum 2 retries**
- **Permanent failure → STOP**
- **Mandate cooldown protection**

If the policy encounters an ambiguous or exceptional condition, the system fails safely rather than automatically executing the action.

12. Mandate Recovery Protection

For recurring payment methods such as NACH, E-Mandate and UPI Autopay, RecoveraX uses a dedicated retry sequencer.

It considers:

- Clearing/presentation windows
- Retry timing
- Customer liquidity timing
- Cooldown requirements

The objective is to avoid repeatedly presenting failed mandates and creating unnecessary dishonor/bounce risk.

13. Human-in-the-Loop

RecoveraX does not attempt to automate every decision.

High-value or risky cases are routed to a human operator.

The operator can review:

- Transaction amount
- Failure diagnosis
- Recovery score
- Recommended action
- Risk indicators
- Relevant case history

This provides a balance between:

Automation for safe cases

and

Human control for risky cases.

14. Verification and Auditability

Every recovery case maintains an auditable lifecycle:

CASE CREATED



DIAGNOSED



SCORED



POLICY EVALUATED



APPROVED / BLOCKED



SCHEDULED



RE-CHECKED



EXECUTED



VERIFIED



FINAL OUTCOME

This provides visibility into:

- What happened
- Why the decision was made
- What action was taken
- Whether a human approved it
- Whether the payment actually succeeded

15. Observability

RecoveraX integrates **LangSmith** for agent tracing and observability.

The system can track:

- Agent execution
- LLM calls
- Workflow steps

- Latency
- Errors
- Recovery outcomes

Sensitive information is sanitized before being sent to the observability layer.

16. Technology Stack



Technologies

Frontend: Next.js

Backend: FastAPI

Agent: LangGraph

LLM: Groq / Qwen

Background Jobs: Celery + Redis

Voice: Sarvam AI

Observability: LangSmith

Safety: Deterministic Python Policy Engine

17. Batch Evaluation

RecoveraX was evaluated using a **1,000-case synthetic payment simulator benchmark**.

The following results are simulator results and are not live Razorpay merchant-production results.

Metric	Baseline: Blind Retry	RecoveraX
Total volume	₹50L	₹50L
Revenue recovered	₹12.5L	₹34.8L
Recovery rate	25.0%	69.6%
Incremental recovery	—	₹22.3L
Duplicate-debit violations	14	0

Key result

RecoveraX recovered **₹34.8L** compared with **₹12.5L** for the baseline, resulting in **₹22.3L** incremental recovery in the synthetic benchmark.

The benchmark also demonstrated the value of safety controls by preventing duplicate-debit scenarios through fresh payment-state verification.

18. Why RecoveraX?

Traditional recovery:

PAYMENT FAILED



RETRY



RETRY



RETRY

RecoveraX:

PAYMENT FAILED



UNDERSTAND WHY



ESTIMATE RECOVERY POTENTIAL



RECOMMEND ACTION



CHECK SAFETY



AUTO / HUMAN / BLOCK



SCHEDULE



RE-CHECK



EXECUTE



VERIFY



RECOVER / RE-EVALUATE / STOP

RecoveraX therefore focuses on **recovering revenue safely, not simply retrying payments more aggressively.**

19. Key Benefits

For Merchants

- Increased recovery of at-risk revenue
- Reduced unnecessary retries
- Protection against duplicate-debit scenarios
- Automation of low-risk cases
- Human control over high-risk cases
- B2B Promise-to-Pay tracking
- Mandate retry optimization
- Batch recovery analytics
- Complete audit trail

For Customers

- Fewer unnecessary payment attempts
- Reduced duplicate-debit risk
- More relevant recovery communication
- Localized Hinglish voice assistance
- Better handling of payment commitments

- Safer payment recovery experience

20. What Makes RecoveraX Different?

RecoveraX is designed around one principle:

Financial AI should be intelligent enough to reason, but constrained enough to be trusted.

The LLM provides reasoning.

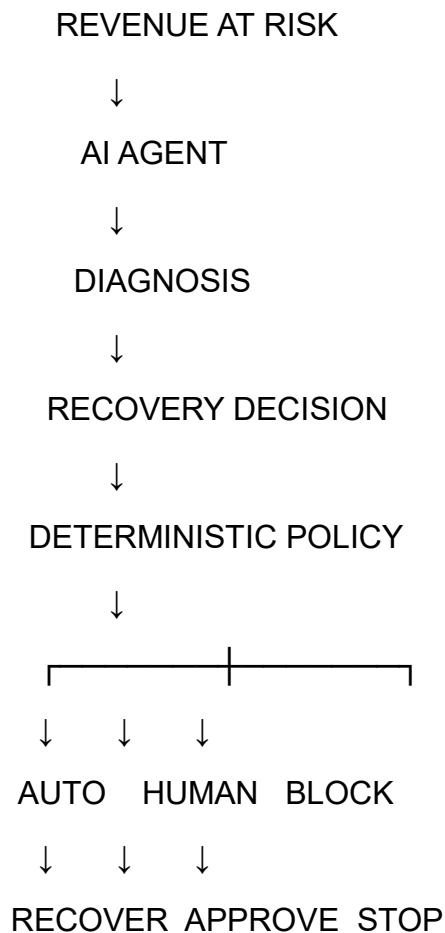
The deterministic policy provides financial authority.

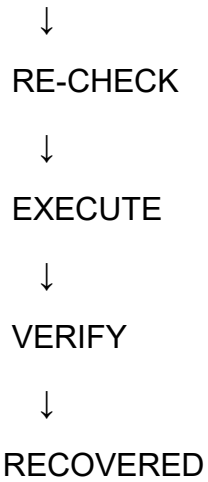
The human provides oversight where required.

The verifier provides proof.

And the batch analytics provide measurable evidence.

Final Architecture

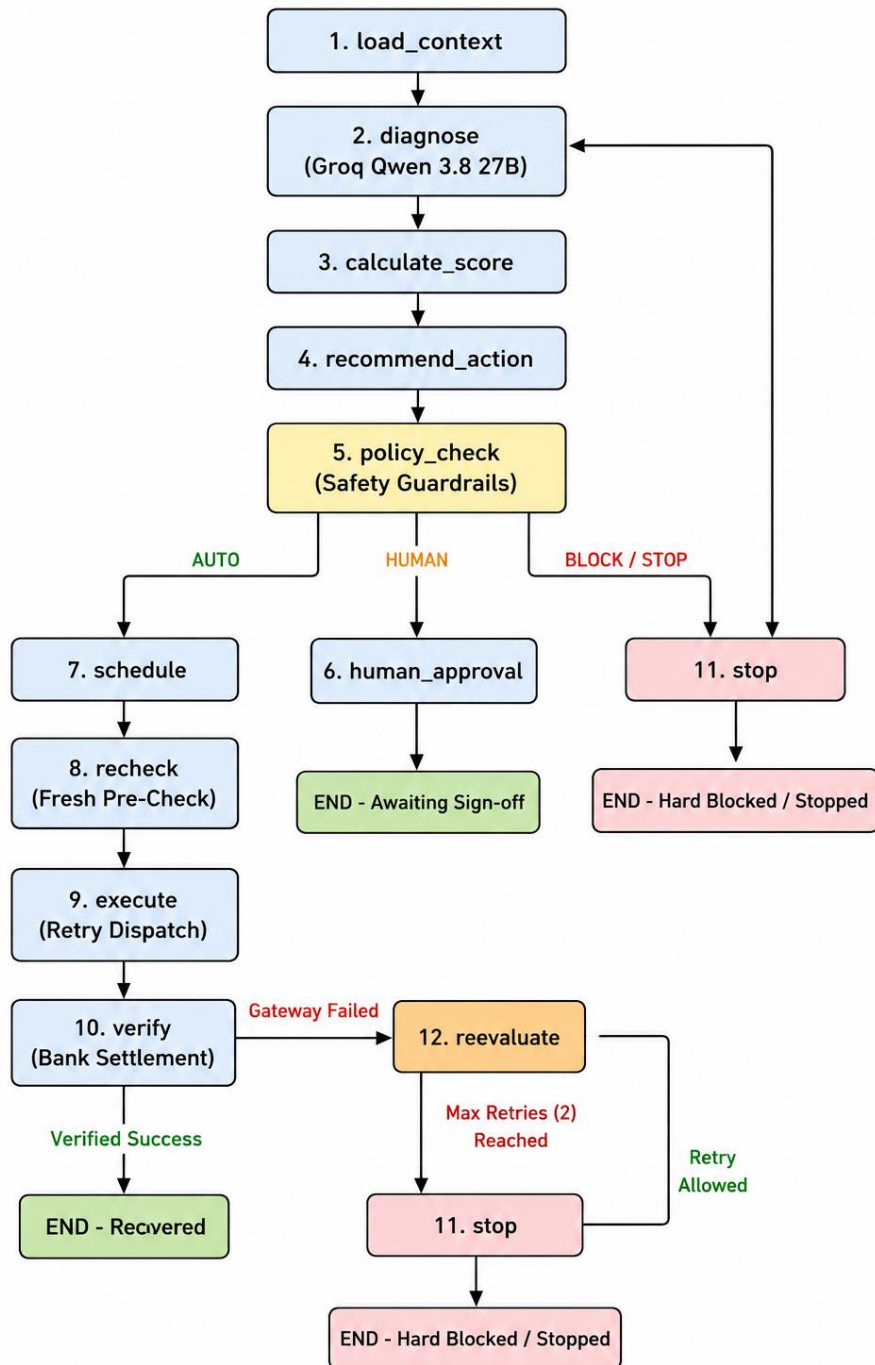




RecoveraX in one sentence

RecoveraX is a safety-first AI revenue recovery agent that turns revenue-loss events into verified recovery through intelligent diagnosis, deterministic financial guardrails, bounded automation, human oversight, and settlement verification.

Langgraph Workflow



LangSmith agent tracing and observability.

Personal / Tracing / RecoveraX

RecoveraX ID Tracing Evaluators Automations Insights Engine

Default View Last 1 day Threads Traces Runs Input Search input

Add filter <>

Name	Input	Output
RecoveraX Recovery Cas...	CASE-1001	CASE-1001
RecoveraX Recovery Cas...	CASE-1001	TEMPORARY_FAILURE
RecoveraX Recovery Cas...	CASE-1005	CASE-1005
RecoveraX Recovery Cas...	CASE-1006	CASE-1006
RecoveraX Recovery Cas...	CASE-1006	CASE-1006
RecoveraX Recovery Cas...	CASE-1005	CASE-1005
RecoveraX Recovery Cas...	CASE-1004	CASE-1004
RecoveraX Recovery Cas...	CASE-1003	AMBIGUOUS_STATE
RecoveraX Recovery Cas...	CASE-1002	CASE-1002
RecoveraX Recovery Cas...	CASE-1001	CASE-1001
RecoveraX Recovery Cas...	CASE-1003	CASE-1003
RecoveraX Recovery Cas...	CASE-1002	CASE-1002
RecoveraX Recovery Cas...	CASE-1001	CASE-1001

Trace

RecoveraX Recovery Case [CASE-1001]

load_context 0.00s

diagnose 0.79s 372

qwen/qwen3.8-27b 0.41s 372

calculate_score 0.00s

recommend_action 0.66s 322

qwen/qwen3.8-27b 0.38s 322

policy_check 0.00s

route_policy_decision 0.00s

schedule 0.00s

Feedback

Input

Fields

- case_id CASE-1001
- amount_at_risk 15000
- audit_events []
- customer
 - id CUST-30018684
 - name Rahul Enterprises
 - average_payment_delay_days 3.3
 - failed_payment_count 0
 - successful_payment_count 9
- enqueue_celery true
- max_retries 2
- retry_count 0
- transaction {10 items}

Output

Personal / Tracing / RecoveraX

RecoveraX ID Tracing Evaluators Automations Insights Engine

Default View Last 1 day Threads Traces Runs Input Search input

Add filter <>

Name	Input	Output
RecoveraX Recovery Cas...	CASE-1005	CUSTOMER_ACTION_REC...
RecoveraX Recovery Cas...	CASE-1001	CASE-1001
RecoveraX Recovery Cas...	CASE-1001	TEMPORARY_FAILURE
RecoveraX Recovery Cas...	CASE-1005	CASE-1005
RecoveraX Recovery Cas...	CASE-1006	CASE-1006
RecoveraX Recovery Cas...	CASE-1006	CASE-1006
RecoveraX Recovery Cas...	CASE-1005	CASE-1005
RecoveraX Recovery Cas...	CASE-1004	CASE-1004
RecoveraX Recovery Cas...	CASE-1003	AMBIGUOUS_STATE
RecoveraX Recovery Cas...	CASE-1002	CASE-1002
RecoveraX Recovery Cas...	CASE-1001	CASE-1001
RecoveraX Recovery Cas...	CASE-1003	CASE-1003
RecoveraX Recovery Cas...	CASE-1002	CASE-1002

Trace

RecoveraX Recovery Case [CASE-1006]

load_context 0.00s

diagnose 3.53s 391

qwen/qwen3.8-27b 0.68s 391

calculate_score 0.00s

recommend_action 2.41s 330

qwen/qwen3.8-27b 0.52s 330

policy_check 0.00s

route_policy_decision 0.00s

human_approval 0.00s

Feedback

Input

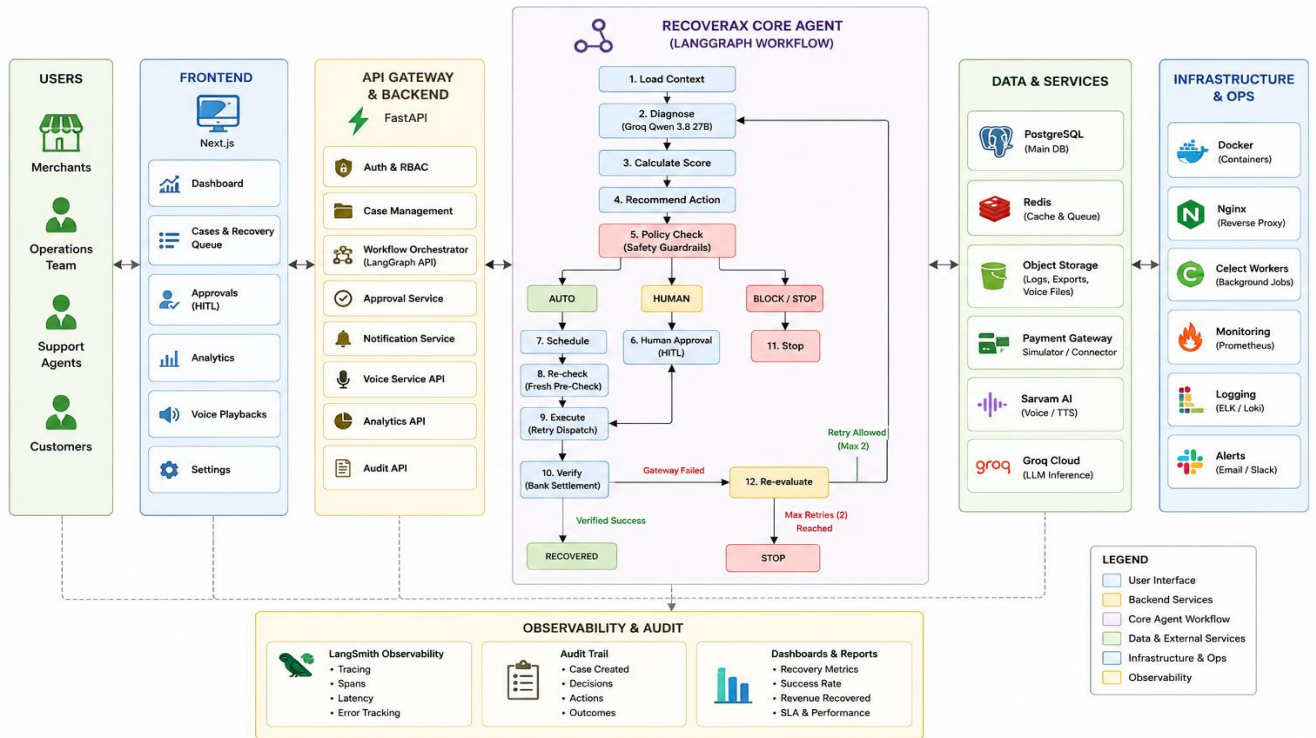
Fields

- case_id CASE-1006
- amount_at_risk 120000
- audit_events []
- customer
 - id CUST-37988176
 - name Global Trade Corp
 - average_payment_delay_days 3.3
 - failed_payment_count 1
 - successful_payment_count 22
- enqueue_celery true
- max_retries 2
- retry_count 0
- transaction {10 items}

Output

RecoveraX Application Architecture

RecoveraX – Application Architecture



About Me

I'm Monishwaran, a passionate software developer and AI enthusiast focused on building practical AI solutions.

Connect

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